

THARUN KUMAR KARNATI

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PROFESSIONAL SUMMARY

Senior Cloud and DevOps Engineer with 9 years of experience designing, automating, securing, and operating scalable cloud infrastructure across AWS and Azure. Skilled in CI/CD, Kubernetes, Docker, Terraform, CloudFormation, Ansible, GitHub Enterprise, Azure DevOps, Jenkins, observability, cloud migrations, DevSecOps practices, and cloud-native application delivery. Experienced in leading enterprise migrations, improving deployment reliability, reducing manual effort, strengthening governance, and mentoring DevOps teams.

TECHNICAL SKILLS

Cloud Platforms: AWS (EKS, ECS, EC2, VPC, RDS, S3, CloudWatch, CloudFront, Elastic Beanstalk, ECR) and Azure (Virtual Network, App Services, Entra ID, Power BI, Microsoft Fabric, Event Hubs, Function Apps, Marketplace, Networking, Service Bus)

AI Tools: ChatGPT, GitHub Copilot CLI, Amazon Q Developer, Claude, Cursor

CI/CD Tools: Jenkins, GitHub Actions, Azure DevOps, Microsoft Visual Studio App Center

Infrastructure as Code: Terraform, CloudFormation, Kubernetes Manifest Files, Ansible

Web Servers: Apache, WildFly, IIS, NGINX, Tomcat

Containers & Orchestration: Docker, Kubernetes, Docker CLI, Kubectl

Monitoring & Observability: Datadog, AWS CloudWatch, Grafana, Prometheus, Loki

Version Control and Build Tools: Git, GitHub, Bitbucket, Maven, Angular CLI, Node.js, JFrog

Scripting and Automation: Shell scripting, PowerShell, AWS CLI

Databases & Caching: MS SQL, MySQL, Redis, Memcached

Ticketing: Jira, Service Now

Frontend & Backend Development: Angular, Node.js

BI & Reporting Tools: Power BI

LMS: Moodle

PROFESSIONAL EXPERIENCE

Tech Blocks | Senior Engineer - L2 | Cloud & DevOps

Dec-2025 to Present

- Architected and deployed **GitHub Enterprise Cloud platform** with **SSO (Okta)** and automated identity lifecycle management via **SailPoint**, enabling secure and scalable developer onboarding.
- Led **enterprise-scale repository migration** from **Azure DevOps (ADO)** and **GitLab** to **GitHub Enterprise**, ensuring 100% preservation of commit history, access controls, and CI/CD configurations.
- Designed and implemented **ServiceNow-GitHub/ADO integration** to automate **Change Management (ITIL)** workflows, improving release governance and audit compliance.
- Developed automation to **capture and synchronize Pull Request (PR) metrics and lifecycle events** from GitHub and ADO into ServiceNow dashboards for real-time visibility.
- Executed **JFrog Artifactory migration and optimization**, including repository restructuring, artifact versioning strategies, and access control standardization.
- Built and standardized **CI/CD pipelines using GitHub Actions and Azure DevOps**, reducing manual deployment effort and improving release consistency across environments.

- Implemented **DevSecOps best practices**, including branch protection rules, mandatory PR reviews, and integration with **SonarQube** for static code analysis and quality gates.
- Automated **artifact publishing and dependency management** using JFrog Artifactory, ensuring traceability, reproducibility, and compliance across releases.
- Drove **DevOps platform adoption across multiple teams**, providing onboarding frameworks, documentation, and hands-on support, resulting in improved developer productivity.
- Established **enterprise DevOps governance standards**, including repository management, access policies, and audit controls aligned with security and compliance requirements.

Innominds | Senior Engineer – Cloud and DevOps

Mar-2023 to Oct -2025

- Reduced **deployment errors by 70%** and improved release consistency by building **CI/CD** pipelines in **Azure DevOps** (YAML) integrated with **Azure Repos**, automating deployment of **Kubernetes manifests** to **AWS EKS**.
 - Improved **microservices scalability** and availability by deploying **15+ services** on **AWS EKS**, using **Ingress controllers** and **AWS external load balancers** to manage secure routing and traffic segmentation.
 - Boosted resource utilization by **25%** by implementing **Horizontal Pod Autoscaling (HPA)**, enabling services to scale **dynamically** based on real-time load and performance metrics.
 - Accelerated recovery and **observability** with real-time dashboards and alerting via Amazon Managed **Prometheus** and **Grafana**, enhancing incident response and reducing MTTR.
 - Cut infrastructure provisioning time by **50%** by automating **EKS cluster** setup, networking, and Kubernetes resource creation with **Terraform**, ensuring repeatable and compliant environments.
 - Standardized **deployment** patterns by maintaining **reusable** Kubernetes **manifests** and **Helm** charts, increasing team velocity and reducing configuration drift across environments.
 - Improved deployment **reliability** and uptime by leveraging **Kubernetes** health checks, rolling updates, and canary deployment strategies to ensure safe rollout of changes.
 - Enabled **secure Power BI integration** by configuring on-premises **data gateways**, allowing seamless access to databases in private networks from Power BI Service.
 - Designed environment-specific **Power BI workspaces** with dedicated **gateway connections**, improving access control and operational efficiency.
 - Automated data pipelines by developing **Azure Serverless** Function Apps, reducing manual effort and cutting processing time by 40%.
 - Built CI/CD pipelines in Azure **DevOps** for automated deployment of **Azure Functions**, ensuring consistent and error-free releases across environments.
 - Provisioned and managed **AWS** infrastructure (EC2, RDS, S3, EKS, etc.) with **Terraform** across multi-environment deployments, ensuring high availability and scalability.
 - Managed **SSL/TLS certificate** renewals for WildFly servers on **Azure VMs**, maintaining secure communication across Dev, QA, and Prod.
 - Scheduled automated **Azure Functions** in Java, executing short-lived data tasks on predefined cron schedules.
 - Configured **self-hosted** DevOps agents in Azure, enabling support for long-running ETL pipelines.
 - Led AWS account **migration**, securely transferring resources (ECR, EC2, S3, EKS, RDS) while maintaining operational continuity.
 - Upgraded **EKS clusters** by analysing Helm chart dependencies and mitigating deprecation risks.
 - Deployed a **GenAI-powered** Python application using a **3-tier Azure architecture**; successfully published as both **SaaS** and **VM offerings** on Azure Marketplace.
 - Migrated repositories and histories from **GitLab** to **GitHub**, streamlining version control, access, and collaboration workflows
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TALKD | Engineer – Cloud and DevOps

Aug-2021 to Jan-2023

- Architected and maintained highly available **AWS** infrastructure including VPCs, EC2, EKS, RDS, Elastic Cache, **Elastic**

Beanstalk, and CloudFront, ensuring 99.9% uptime and **scalability** for critical business applications.

- Designed and **automated** infrastructure provisioning using **Terraform**, integrating **ECS** services with **Jenkins**-based CI/CD pipelines to accelerate application delivery.
- Led cloud **cost optimization** initiatives, preparing detailed reports and presentations, and collaborating in audits to identify **cost-saving** opportunities, achieving measurable reduction in monthly cloud spend.
- Executed **zero-downtime migration** of critical **databases** from SQL Server Enterprise Edition to Web Edition, maintaining data integrity and uninterrupted service availability.
- Spearheaded application modernization by **migrating monolithic** applications to **microservices** architecture, improving scalability and deployment agility.
- **Led Jenkins migration and upgrade initiatives**, resolving plugin compatibility issues, integrating **SonarQube** for code quality checks, linking **Jira** for issue tracking, and setting up automated email alerts for build and deployment failures—resulting in improved pipeline stability and faster issue resolution.
- Led **Jenkins migration** projects, creating pipeline scripts and freestyle jobs to automate build and **deployment** workflows, improving deployment frequency and **reliability**.
- Managed **mobile** application **CI/CD** pipelines, deploying apps via **Microsoft** App Center to Google Play Store and Apple App Store, enhancing release cadence and app quality.
- Monitored and troubleshoot **Jenkins** build pipelines, analyzing logs and resolving failures, reducing build downtime by 25%.
- Oversaw Elastic **Beanstalk** deployments, validating logs and events to ensure smooth and successful production releases.
- Scaled **ECS** clusters, **Redis**, **Memcached**, RDS, and **EC2** instances to meet performance testing and production traffic demands, ensuring system responsiveness under load.
- Built custom monitoring dashboards and alerts in **Datadog**, integrating AWS **CloudWatch** metrics with **autoscaling** events to proactively manage infrastructure health.
- Administered **GitHub** Enterprise repositories, implementing robust branching, merging, and tagging strategies to streamline development workflows and version control.

ICROZ | Engineer – Cloud and DevOps

Mar-2017 to Jul-2021

- **Ensured 99.95% uptime during multiple product launches** by proactively monitoring servers, managing incidents, and coordinating rapid responses using Azure Monitor and CloudWatch.
- **Automated end-to-end CI/CD pipelines** leveraging Jenkins, Maven, Nexus, and Ansible on Linux platforms, accelerating deployment frequency by 35%.
- **Architected and maintained Jenkins master-slave clusters**, enabling parallel build execution and reducing build times by 40%.
- **Designed and implemented hub-and-spoke VNets** for secure and scalable multi-environment Azure architecture (DEV, QA, PROD), improving network isolation and compliance.
- **Configured Azure Network Security Groups (NSGs) and Azure Firewall policies** to tightly control traffic flow, achieving compliance with organizational security standards.
- **Monitored and optimized Azure Load Balancer performance** using Azure Monitor and Log Analytics, ensuring SLA compliance and minimizing service interruptions.
- **Implemented Blob Storage lifecycle management policies**, automating data tiering between Hot, Cool, and Archive tiers, resulting in a 25% reduction in storage costs.
- **Deployed Azure Load Balancer with health probes**, ensuring only healthy backend VM instances serve traffic and reducing downtime by 30%.
- **Implemented auto-scaling and load balancing mechanisms**, optimizing resource utilization and improving cost efficiency across cloud environments.
- **Secured PaaS service access** by deploying private endpoints for Storage, SQL Database, and Key Vault, enhancing data security and compliance.
- **Enhanced network visibility and troubleshooting** using Azure Network Watcher and Traffic Analytics, decreasing incident resolution times by 20%.
- **Monitored AWS server metrics via CloudWatch**, setting up alarms and retention policies that proactively identified performance bottlenecks.
- **Developed and maintained Jenkins CI jobs for microservices**, automating artifact deployment to Apache Tomcat servers and improving release reliability.
- **Automated AWS infrastructure provisioning** using Ansible playbooks to deploy and manage EC2 instances and EBS

volumes, reducing manual setup time by 50%.

- **Designed highly available and fault-tolerant AWS VPC architectures** with public/private subnets across multiple Availability Zones, improving system resilience and security posture.

EDUCATION

- Bachelor of Technology in Mechanical Engineering (2016, JNTU-Hyderabad).